Stromversorgung

File: power.kicad_sch

Steuerung_ESP32S3

File: ctrl_s3.kicad_sch

HMI_ESP32P4

File: hmi_p4.kicad_sch

Analogeingänge

File: analog_in.kicad_sch

Analogausgänge

File: analog_out.kicad_sch

Digital_RS485_Luefter

File: io.kicad_sch

Pumpensteuerung — AIO-Steuerplatine, Rev 0.2 (Entwurf)
Ersetzt LOGO! 8.4 + AM2 + Raspberry Pi 3B+. Zwei Controller ohne Linux:
  ESP32-S3 = Steuerung (Sensoren, PI-Druckregelung, V20 per RS485, Relais, AO, Trockenlaufschutz)
  ESP32-P4 (M5Stack Stamp-P4) = App/HMI (Web-UI, Bewässerung, Wetter, MQTT/HA, 7"-DSI-Touch, LAN)
  Verbindung: UART (S3-UART0 ↔ P4), P4 kann S3 resetten/flashen.
Versorgung 230 V AC → IRM-30-24 → 24 V → 12 V (Lüfter) / 5 V (P4, Display) → 3,3 V (S3, Logik).
Details: hardware/SPEC.md

Erzeugt mit hardware/tools/gen_schematic.py

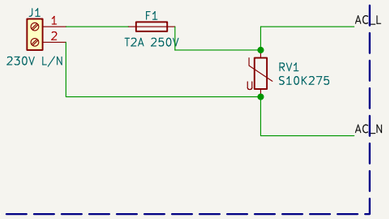
Brunnensteuerung Wathlingen

Sheet: /
File: pumpensteuerung.kicad_sch

Title: Pumpensteuerung — Übersicht

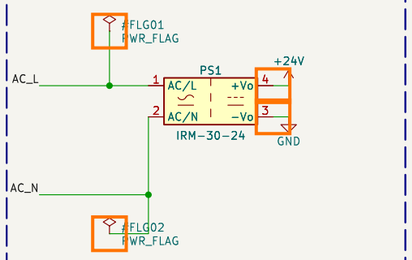
Size: A3	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/7

Netzeingang 230 V AC

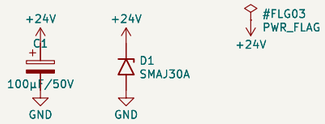


F1 T2A träge (5×20) vor dem Netzteil, RV1 275 VAC gegen Netzspitzen.
Netzbereich: ≥ 6 mm Kriechstrecke + Frässchlitz zur Kleinspannung.

Netzteil 24 V / 30 W

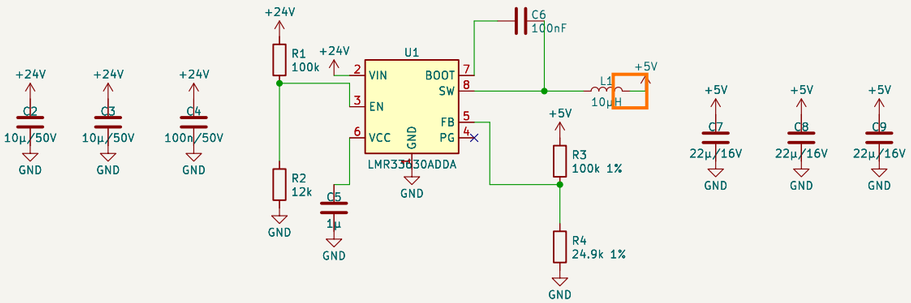


24-V-Schiene



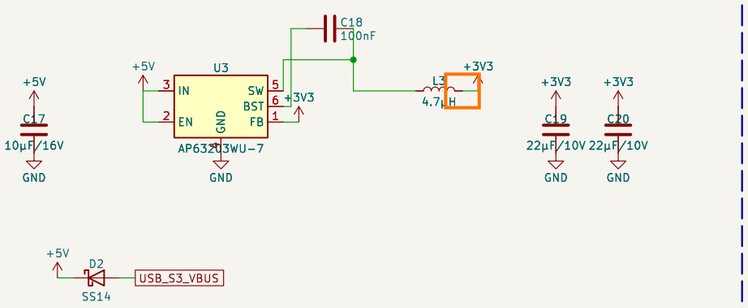
C1 Pufferung, D1 SMAJ30A gegen Überspannung auf 24 V.

24 V → 5 V / 3 A (Stamp-P4 + 7"-Display)



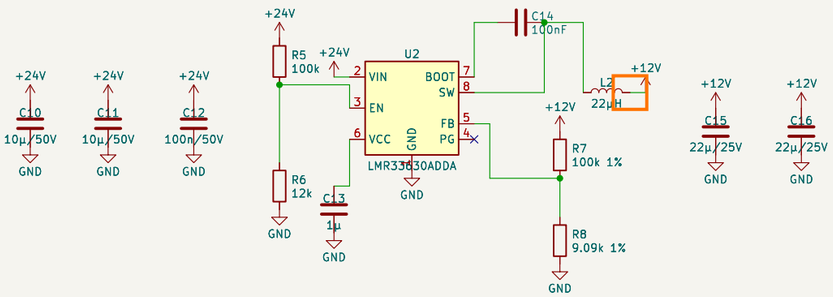
LMR33630 400 kHz. U = 1 V · (1 + 100k / 24,9k) = 5,0 V. EN-Teiler: Einschalten ab ≈ 11 V.

5 V → 3,3 V / 2 A



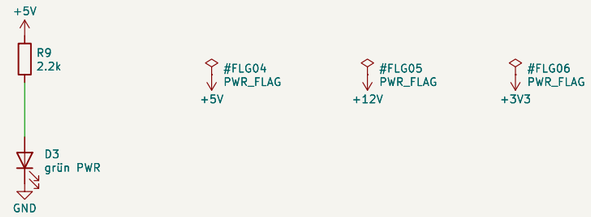
AP63203 (Festspannung 3,3 V) für ESP32-S3, ADS1115, MCP4728, ADM2587E, LAN8720A.
D2 SS14: USB-C des S3 versorgt die Platine auf der Werkbank (ohne 230 V).

24 V → 12 V (Gehäuselüfter)



U = 1 V · (1 + 100k / 9,09k) = 12,0 V.

Betriebsanzeige & Power-Flags



Erzeugt mit hardware/tools/gen_schematic.py
230 V AC → IRM-30-24 → 24 V; daraus 12 V (Lüfter), 5 V (P4, Display), 3,3 V (S3, Logik)

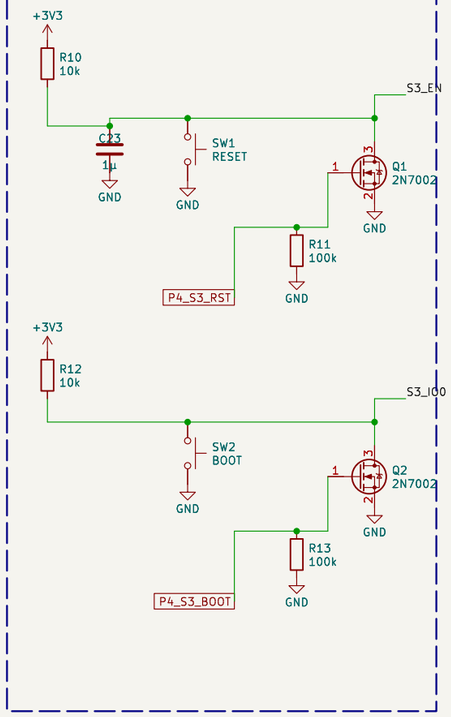
Brunnensteuerung Wathlingen

Sheet: /Stromversorgung/
File: power.kicad_sch

Title: Stromversorgung 230 V / 24 V / 12 V / 5 V / 3,3 V

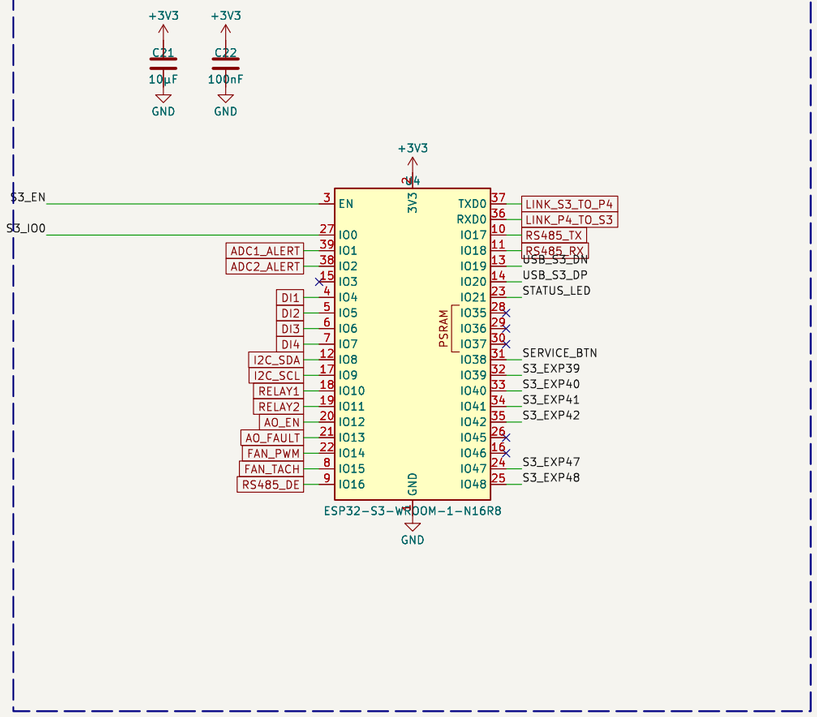
Size: A3	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/7

Reset / Boot (Taster + vom P4 steuerbar)



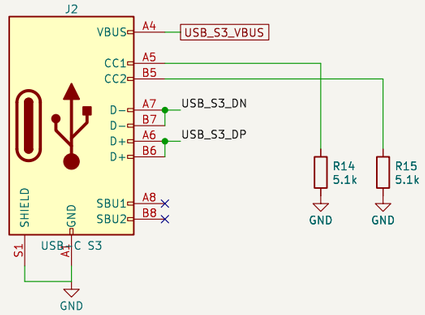
P4_S3_RST / P4_S3_BOOT = HIGH zieht EN bzw. IO0 auf GND.

ESP32-S3-WROOM-1-N16R8

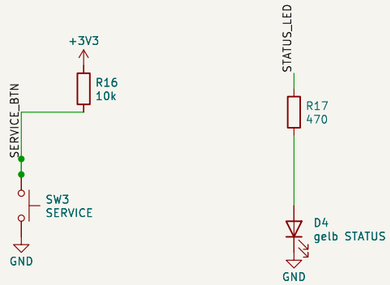


IO3/IO45/IO46 = Strapping, IO35-37 = Octal-PSRAM → frei.
UART0 (RXD0/TXD0) = Link zum P4 und zugleich Flash-Port → P4 kann den S3 per OTA flashen.

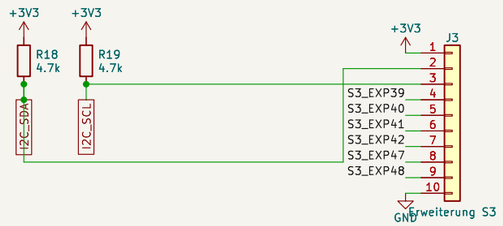
USB-C (Flashen, Konsole, Werkbank-Versorgung)



Bedienung & Status



I²C-Pull-ups & Erweiterung

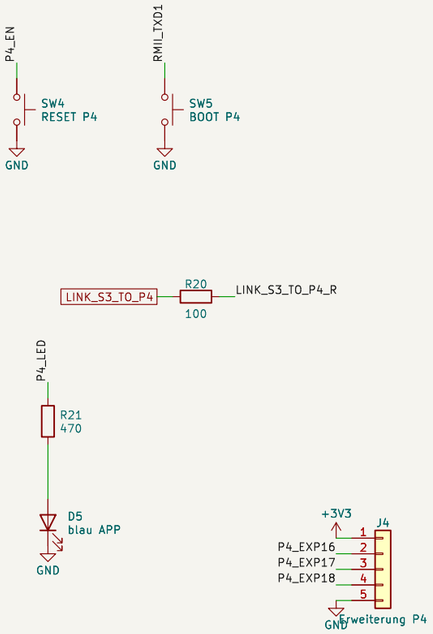


Erzeugt mit hardware/tools/gen_schematic.py
Regelung, I/O, V20 per RS485, Sicherheit – läuft autark
Brunnensteuerung Wathlingen
Sheet: /Steuerung_ESP32S3/
File: ctrl_s3.kicad_sch

Title: Steuerung: ESP32-S3-WROOM-1

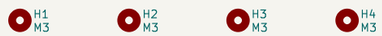
Size: A3	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7

Bedienung, Link zum S3, Erweiterung

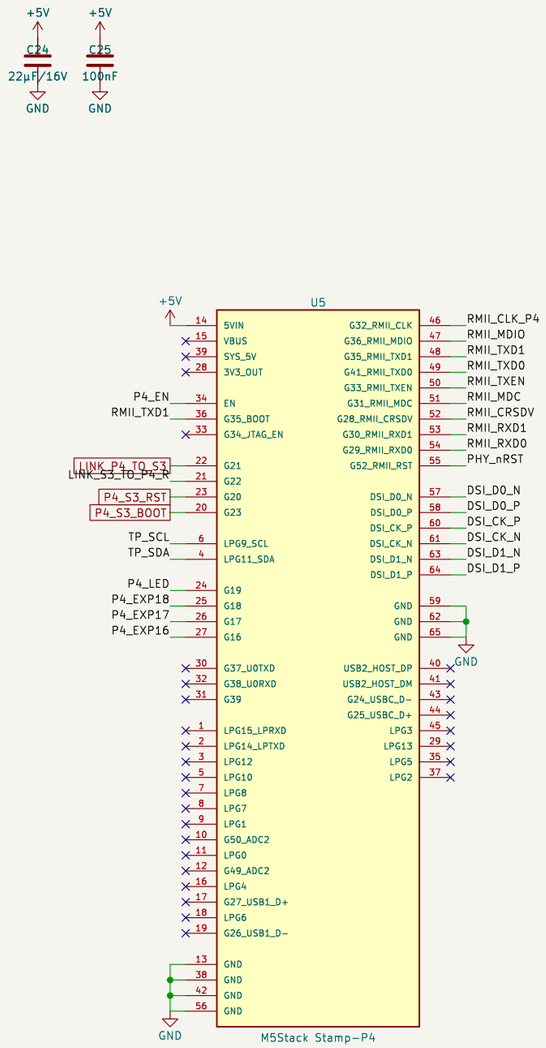


BOOT P4 liegt auf G35 (= RMII_TXD1), nur beim Reset drücken.

Befestigung

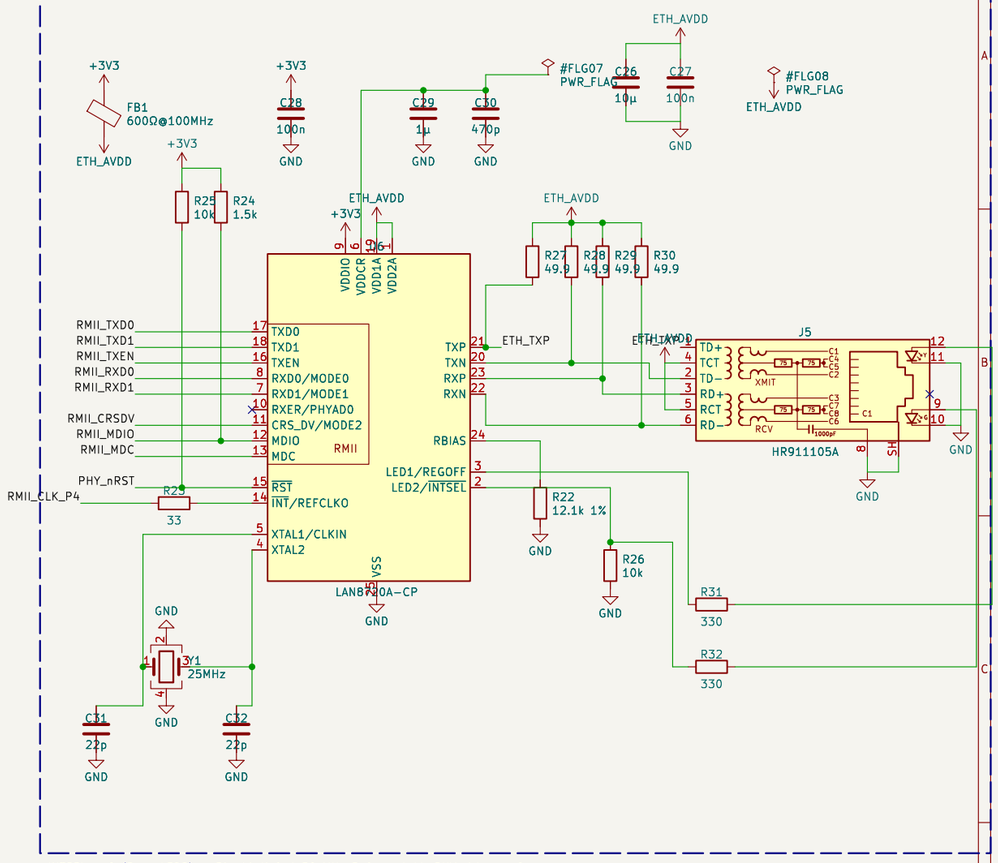


M5Stack Stamp-P4 (ESP32-P4, 16 MB Flash, 32 MB PSRAM)



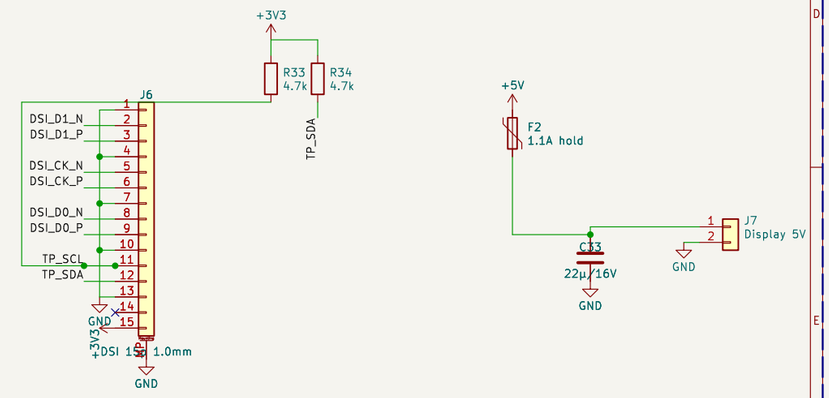
Modul wird aufgelötet (RMII + DSI nur an den Randpads oben).
Eigener USB-C des Moduls = Flash/Konsole P4 → an den Platinenrand setzen.

Ethernet 10/100 (LAN8720A, 25-MHz-Quarz, REFCLKO 50 MHz → P4)



nINTSEL = 0 (R an LED2) → Pin 14 liefert 50 MHz Referenztakt. PHY-Adresse 0.
4 × 49,9 Ω an ETH_AVDD (gefiltert über FB1), Mittelanzapfungen ebenfalls an ETH_AVDD.

7"-Display (MIPI-DSI 2-Lane, FPC 15-pol. 1,0 mm, Raspberry-Pi-Belegung)



Touch-I²C über FPC (Pin 11/12). Display-Logik 3,3 V über Pin 15, Hintergrundbeleuchtung 5 V separat.
DSI-Paare 100 Ω differentiell, so kurz wie möglich.

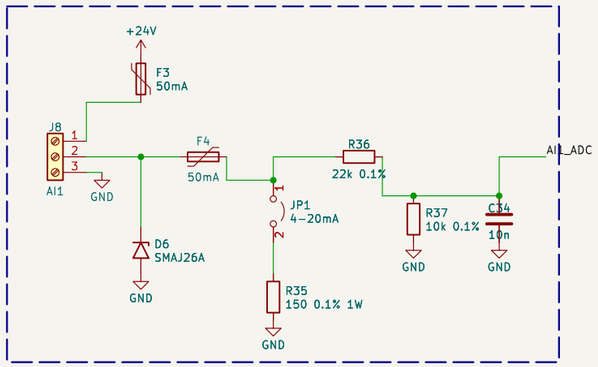
Erzeugt mit hardware/tools/gen_schematic.py
Web-UI, Bewässerung, Wetter, MQTT/HA, LVGL-Touch
Brunnensteuerung Wathlingen

Sheet: /HMI_ESP32P4/
File: hmi_p4.kicad_sch

Title: App/HMI: Stamp-P4, Ethernet, 7"-Display

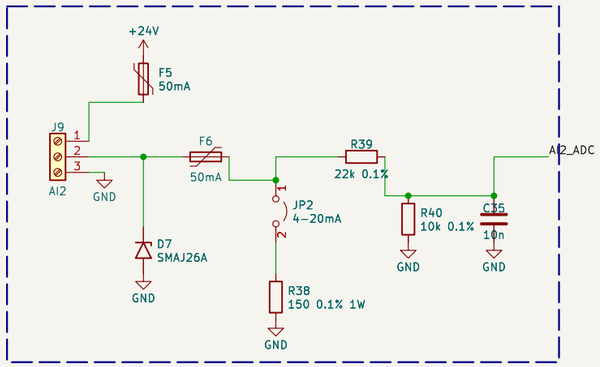
Size: A3	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7

AI1 – Druck 0-10 V

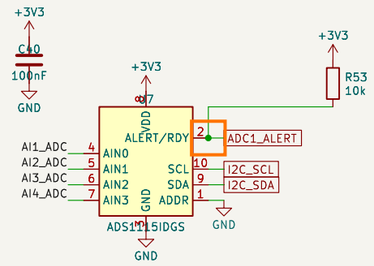


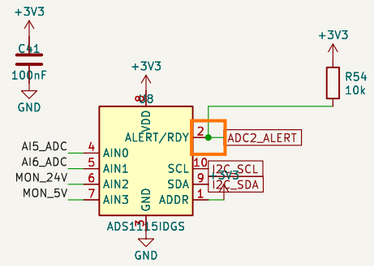
JP gesteckt: 4-20 mA → 150 Ω. JP offen: 0-10 V.
PTC 50 mA in Speisung und Eingang, TVS SMAJ26A.

AI2 – Durchfluss 4-20 mA



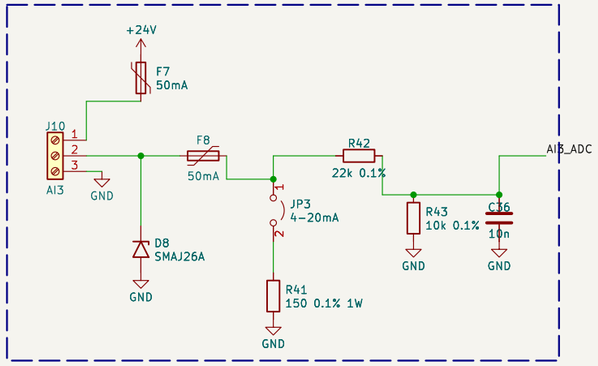
ADC ADS1115 × 2 (I²C 0x48 / 0x49)



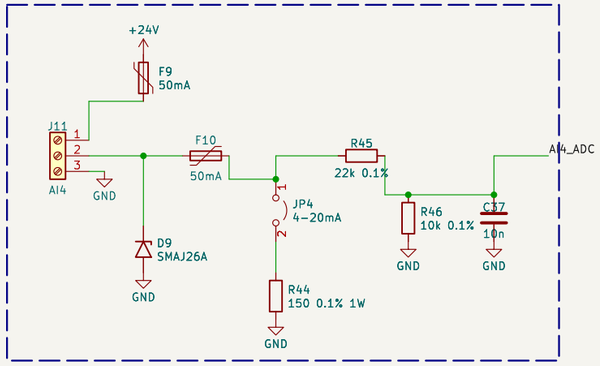


ADC1: AI1-AI4. ADC2: AI5, AI6, 24-V- und 5-V-Überwachung.
4-20 mA: PGA ±1,024 V (20 mA → 0,9375 V). 0-10 V: PGA ±4,096 V (10 V → 3,125 V).

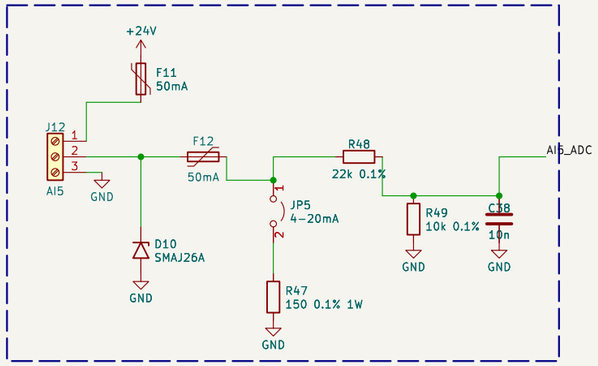
AI3 – Wassertemp 4-20 mA



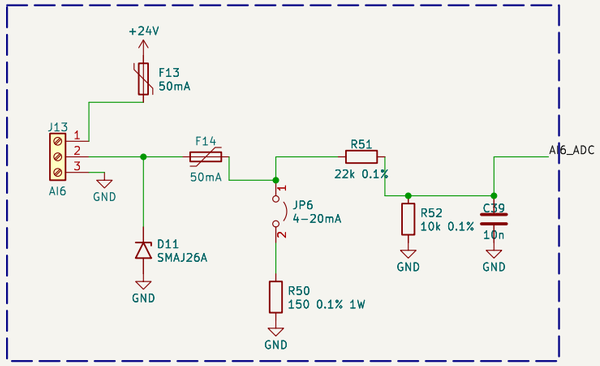
AI4 – Regentonne 4-20 mA



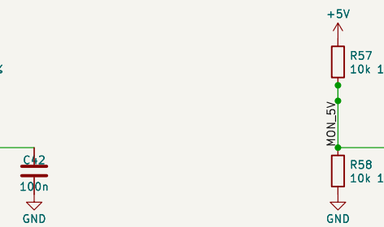
AI5 – Reserve



AI6 – Reserve



Versorgungsüberwachung

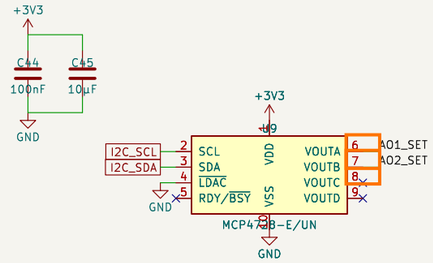


MON_24V = 24 V / 11 (2,18 V). MON_5V = 5 V / 2 (2,5 V).

Erzeugt mit hardware/tools/gen_schematic.py
Jumper gesteckt = 4-20 mA (150 Ω), offen = 0-10 V; Teiler 22k/10k → ADS1115
Brunnensteuerung Wathlingen

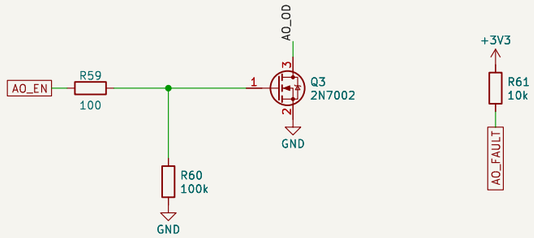
Sheet: /Analogeingänge/		
File: analog_in.kicad_sch		
Title: Analogeingänge 6 × 0-10 V / 4-20 mA		
Size: A3	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 5/7

DAC MCP4728 (I²C 0x60, int. Ref. 2,048 V)



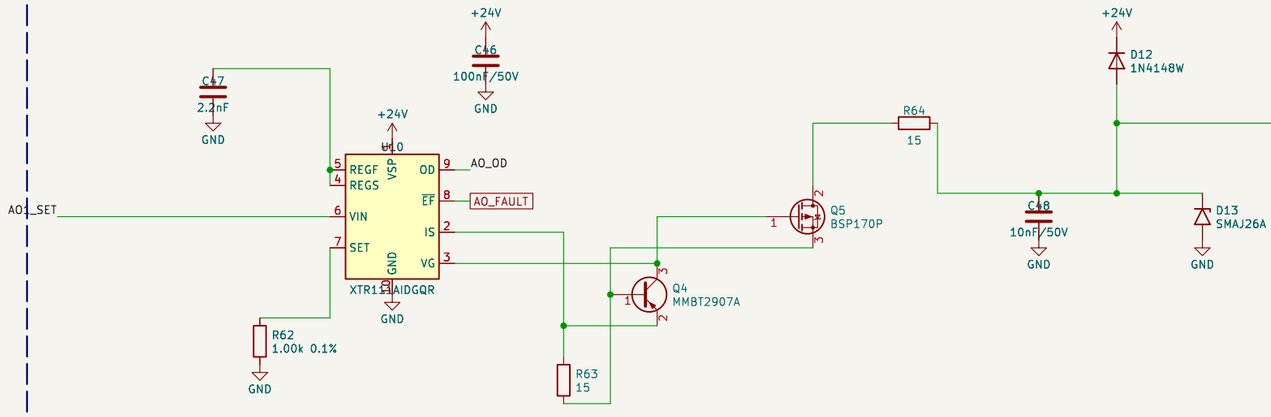
4 mA = Code 800, 20 mA = Code 4000.

Ausgangsfreigabe & Fehlermeldung



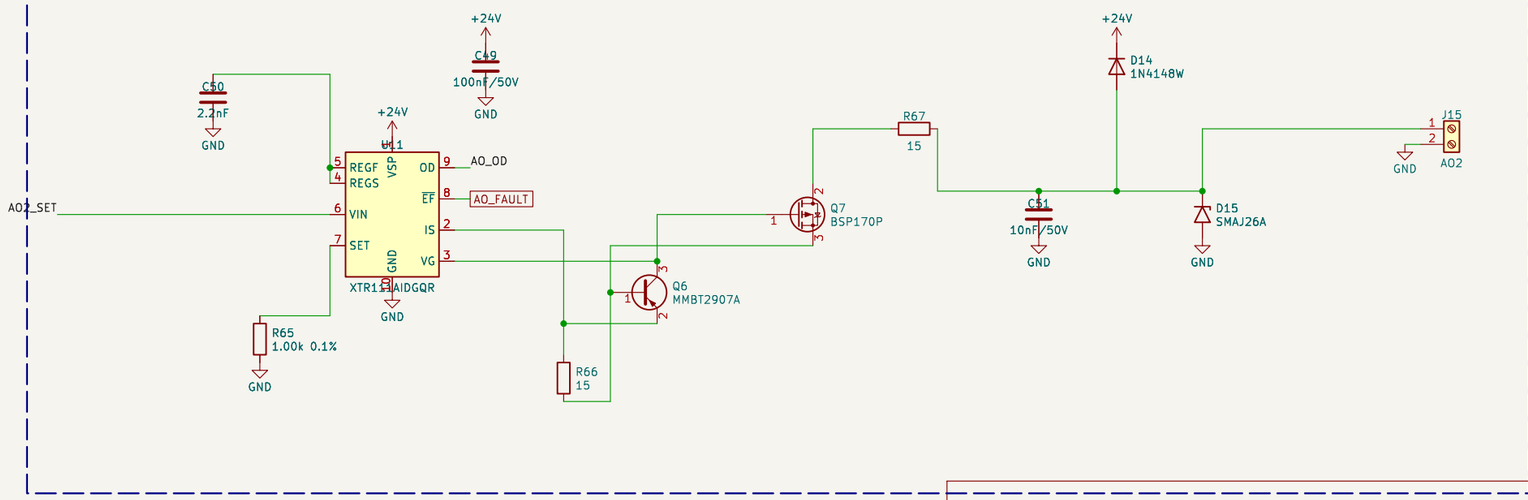
AO_EN LOW (Boot, Absturz) → OD = HIGH → beide Ausgänge 0 mA.
AO_FAULT LOW = Drahtbruch / Bürde zu hoch.

AO1 – Dosierpumpe (0...20,48 mA, Bürde ≤ 900 Ω)



PNP + 15 Ω begrenzt auf ≈ 37 mA. Diode gegen Überspannung an +24 V, TVS gegen GND.

AO2 – Reserve (0...20,48 mA, Bürde ≤ 900 Ω)



Erzeugt mit hardware/tools/gen_schematic.py
MCP4728 → XTR111 → BSP170P; I = 10 · U_DAC / 1 kΩ

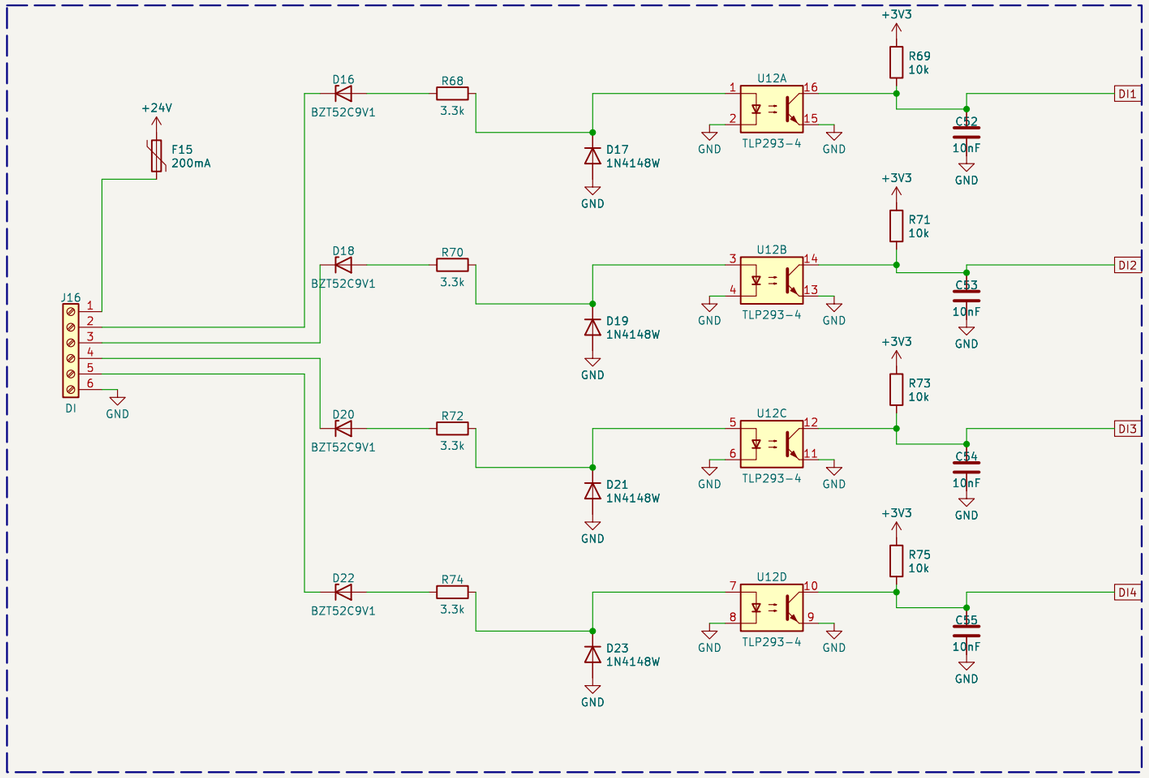
Brunnensteuerung Wathlingen

Sheet: /Analogausgänge/
File: analog_out.kicad_sch

Title: Analogausgänge 2 × 0/4-20 mA

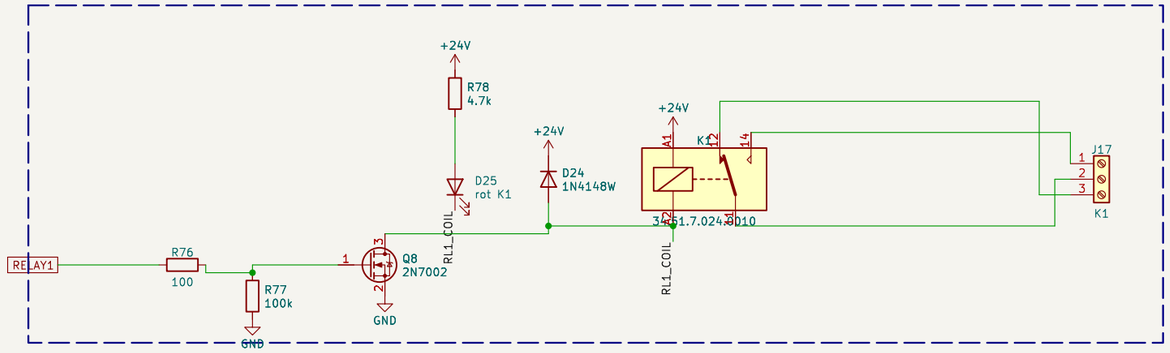
Size: A3	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/7

Digitaleingänge 24 V (aktiv = GPIO LOW)

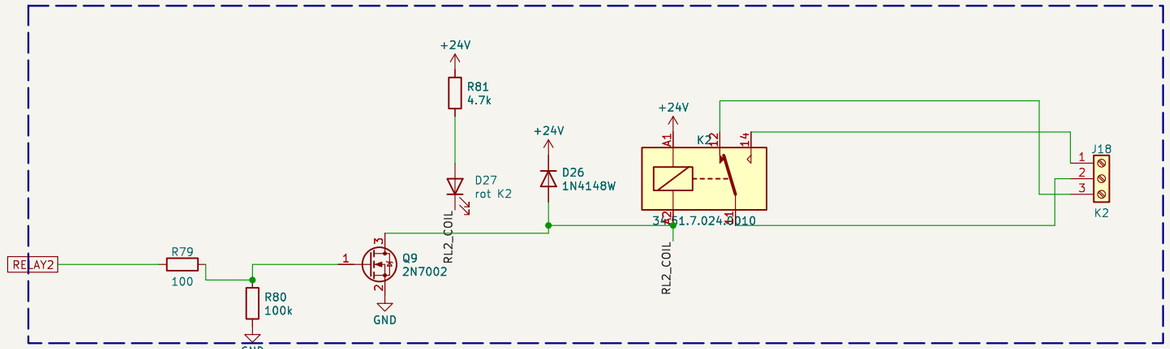


Schaltschwelle ≈ 12 V (Z-Diode 9,1 V + LED). PTC 200 mA schützt die DI-Speisung.

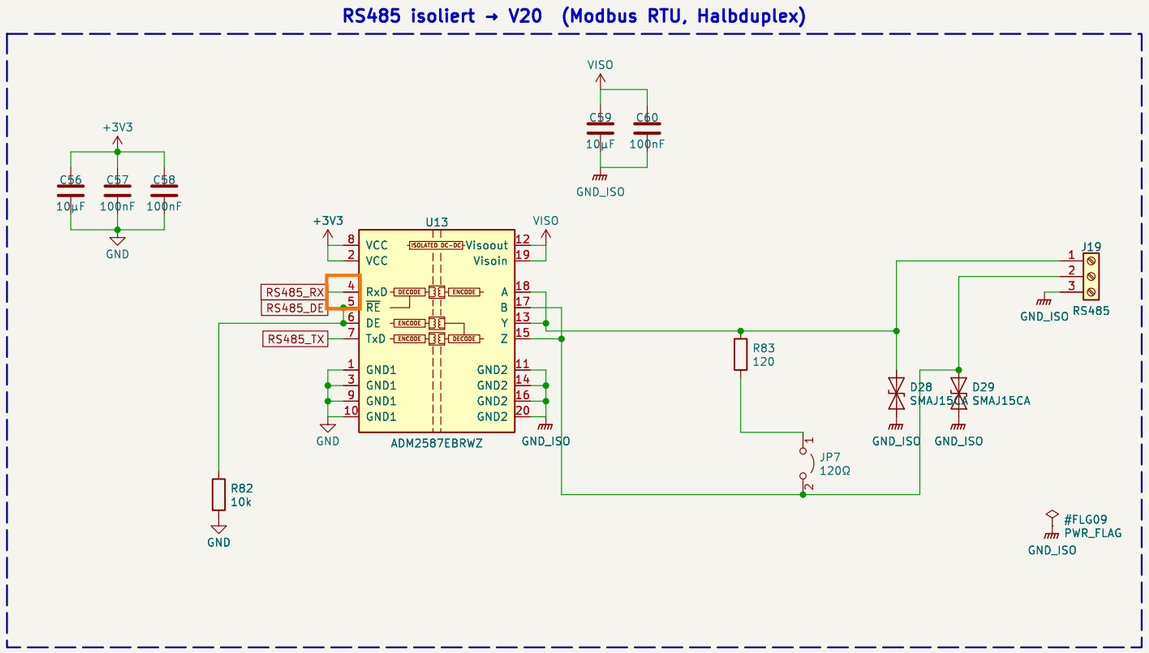
Relais K1 (230 V / 6 A, potentialfrei) — z. B. Freigabe Dosierpumpe



Relais K2 (230 V / 6 A, potentialfrei)

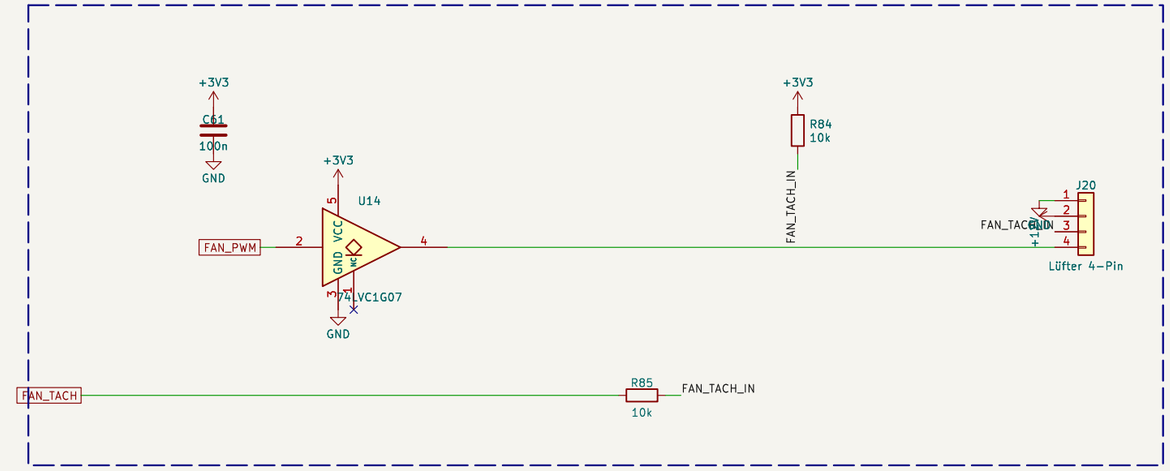


RS485 isoliert → V20 (Modbus RTU, Halbduplex)



DE + /RE gemeinsam (RTS/GPIO16). 10k Pull-down = Empfang. 120 Ω per Jumper.
Klemme: A = P+, B = N-. GND_ISO an V20-Bezugspotenzial.

Gehäuselüfter 12 V (4-Pin-PWM 25 kHz, Tacho)



74LVC1G07 Open-Drain: LOW = Lüfter langsam, offen = Lüfter-interner Pull-up → voll.
Tacho mit Pull-up auf 3,3 V und 10k Serienwiderstand.

Erzeugt mit hardware/tools/gen_schematic.py
DI 24 V über TLP293-4, Relais 230 V, RS485 isoliert zum V20, 4-Pin-PWM-Lüfter
Brunnensteuerung Wathlingen

Sheet: /Digital_RS485_Luefter/
File: io.kicad_sch

Title: Digitaleingänge, Relais, RS485, Lüfter

Size: A3	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 7/7